Rabina Twayana | 12413887 | 06.01.2025

Assignment: Sen2Cube

1. Background

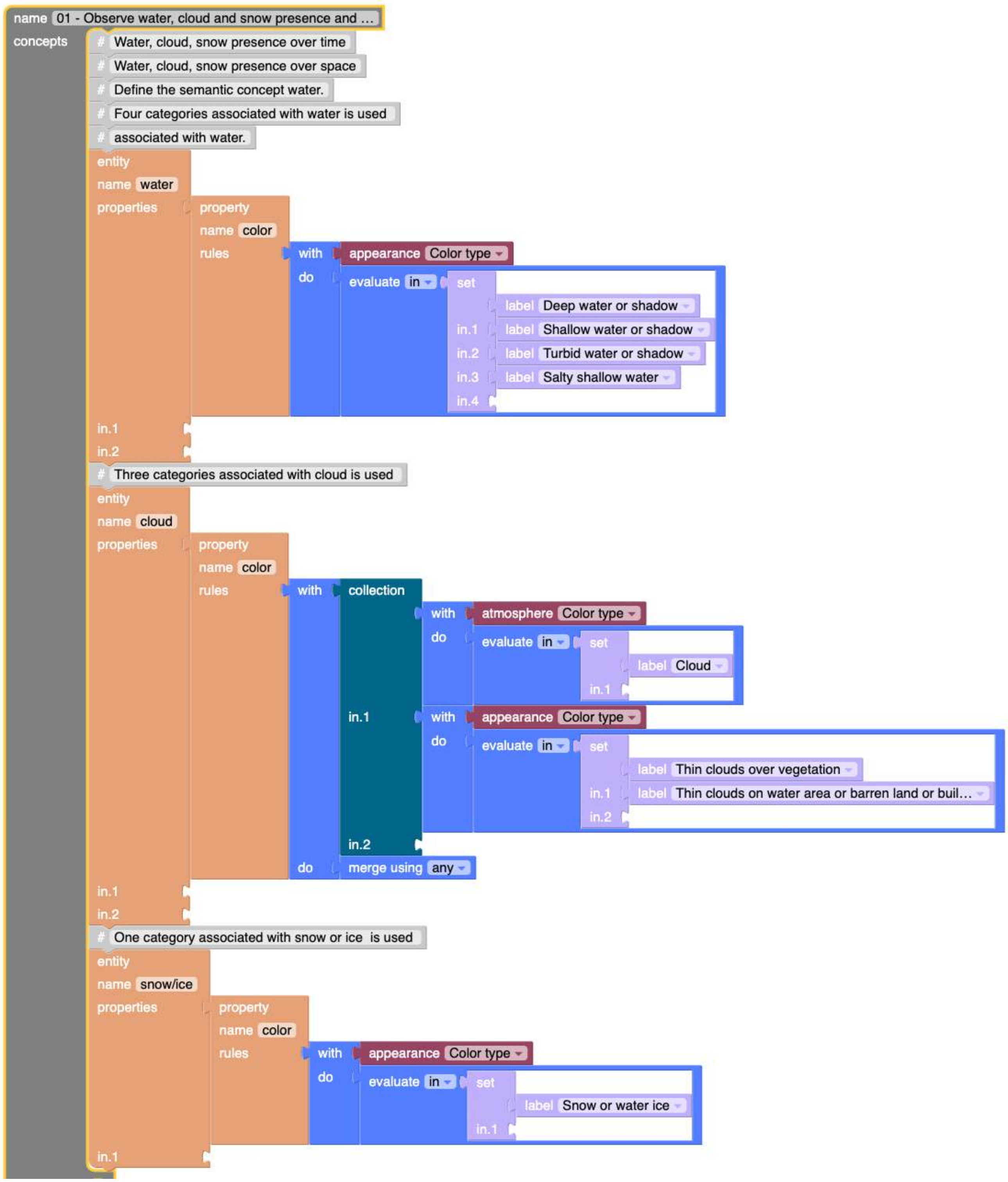
Lakes in Austria have different characteristics throughout the year. They might be snow-covered, iced-covered, or increase/decrease in water level at different times. Thus, the main aim of this task is to visualize the cloud coverage percentage, snow/ice coverage percentage and compare it with the water content by developing a semantic model in Sen2Cube web application. Sentinel 2 Band 2 reflectance value visualization is also added to the model to support the analysis.

The analysis is conducted for the year 2023 focussing on three distinct areas of interest: Traunsee Lake, Millstatter Lake, and Neusiedl Lake which are in the Northern, Southern, and Eastern parts of Austria respectively.

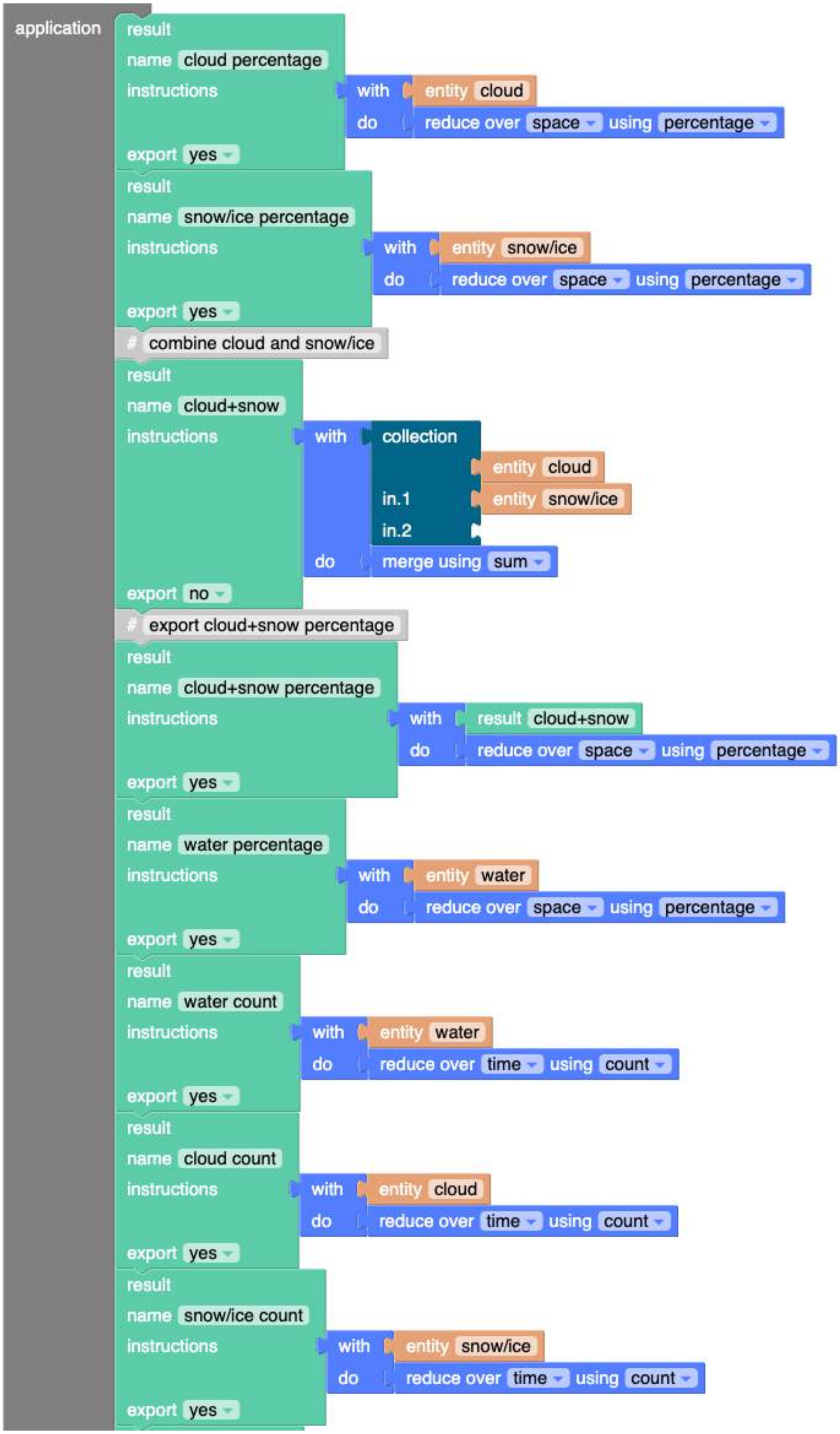
2. Model Development in Sen2Cube

A semantic model represented in Figure 1 below was developed to analyze the presence of water, clouds, and snow, along with the average values of Sentinel-2 Band 2 reflectance (both with and, without cloud and snow/ice cover). Initially, water, cloud, and snow/ice entities were defined. **Water entity** is defined with color type appearance property that includes four classes: deep water or shadow, shallow water or shadow, turbid water or shadow, and salty shallow water. **Cloud entity** is defined by atmospheric cloud property and appearance properties (thin cloud over vegetation and thin cloud on water area or barren land or built-up areas. Finally, **snow/ice entity** is defined with snow/ice appearance property.

In the result section, **cloud percentage, snow/ice percentage, cloud+snow/ice percentage** and **water percentage** blocks are added to visualize cloud, snow/ice, cloud+snow/ice and water percentage coverage reduced over space respectively. Furthermore, **water count, cloud count** and **snow/ice count** result blocks are defined to visualize the water, cloud and snow/ice count reduced over time respectively. Finally, **s2 b2 mean value** block is defined to visualize the average reflectance value reduced over space. Likewise, **s2 b2 cloud+snow free mean value** block is defined to visualize cloud+snow free band 2 reflectance value both in space and time reduction.


name 01 - Observe water, cloud and snow presence and ...
concepts
Water, cloud, snow presence over time
Water, cloud, snow presence over space
Define the semantic concept water.
Four categories associated with water is used
associated with water.
entity
name water
properties
property
name color
rules
with
appearance Color type
do
evaluate in
set
label Deep water or shadow
in.1
label Shallow water or shadow
in.2
label Turbid water or shadow
in.3
label Salty shallow water
in.4
in.1
in.2
Three categories associated with cloud is used
entity
name cloud
properties
property
name color
rules
with
collection
with
atmosphere Color type
do
evaluate in
set
label Cloud
in.1
in.1
with
appearance Color type
do
evaluate in
set
label Thin clouds over vegetation
in.1
label Thin clouds on water area or barren land or buil...
in.2
in.2
do
merge using any
in.1
in.2
One category associated with snow or ice is used
entity
name snow/ice
properties
property
name color
rules
with
appearance Color type
do
evaluate in
set
label Snow or water ice
in.1
in.1

cont...


application
result
name cloud percentage
instructions
with entity cloud
do reduce over space using percentage
export yes
result
name snow/ice percentage
instructions
with entity snow/ice
do reduce over space using percentage
export yes
combine cloud and snow/ice
result
name cloud+snow
instructions
with collection
entity cloud
in.1 entity snow/ice
in.2
do merge using sum
export no
export cloud+snow percentage
result
name cloud+snow percentage
instructions
with result cloud+snow
do reduce over space using percentage
export yes
result
name water percentage
instructions
with entity water
do reduce over space using percentage
export yes
result
name water count
instructions
with entity water
do reduce over time using count
export yes
result
name cloud count
instructions
with entity cloud
do reduce over time using count
export yes
result
name snow/ice count
instructions
with entity snow/ice
do reduce over time using count
export yes

cont..

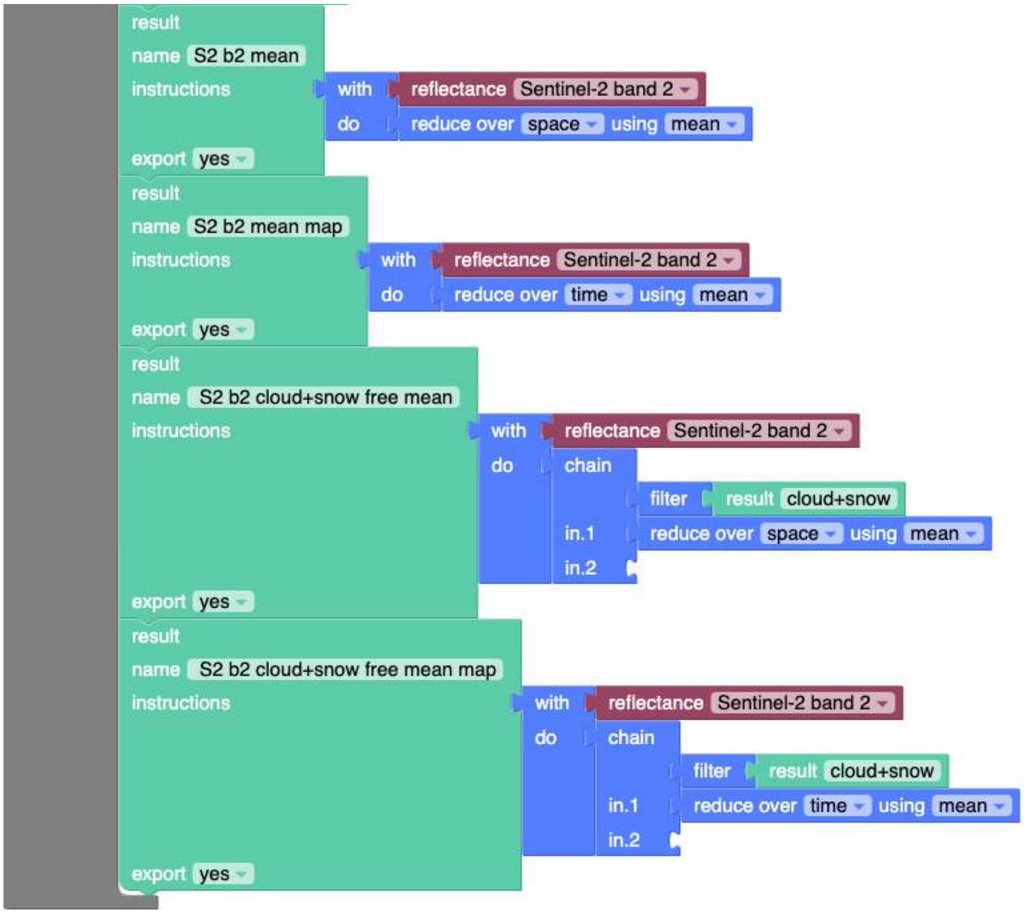


Figure 1: Sen2Cube Semantic Model

3. Results and Discussion

3.1 Traunsee Lake

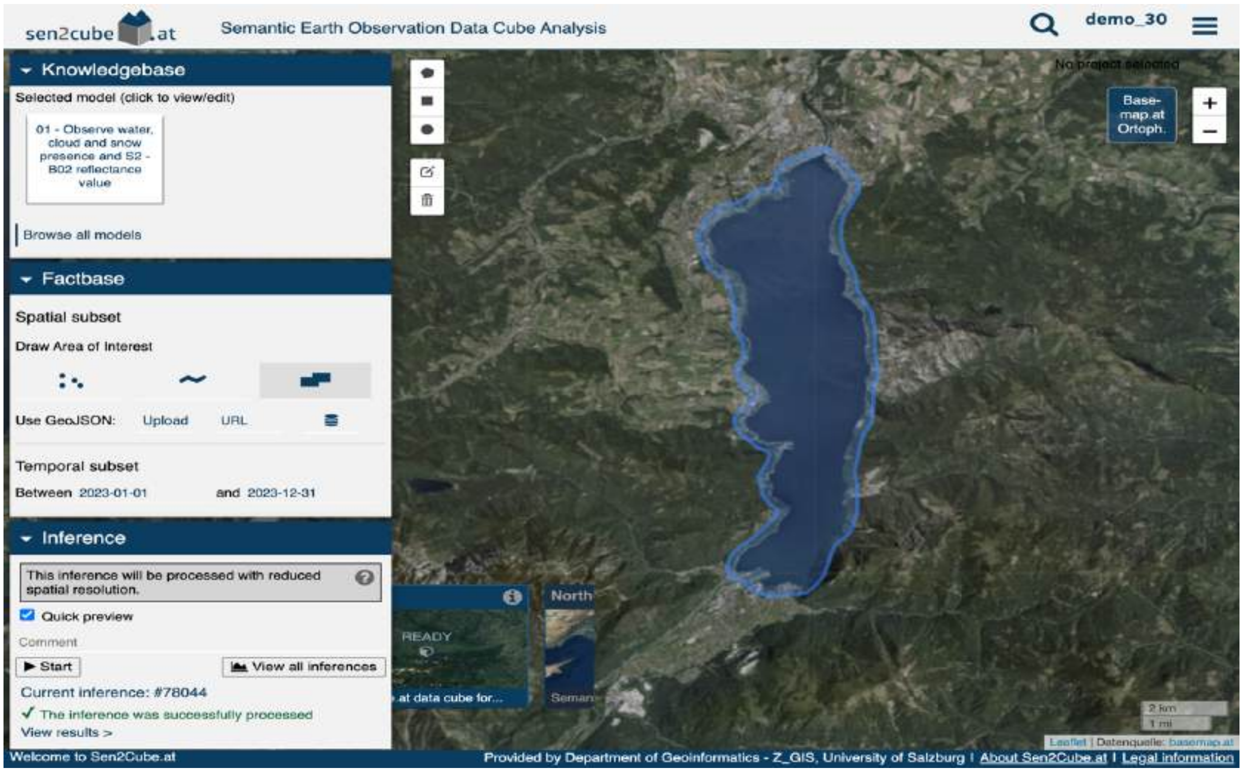


Figure 2: Traunsee lake study area inference setup

Figure 2 represents the Traunsee Lake study area map which is drawn freehand using a polygon draw tool. Model were developed and selected in knowledgebase, spatial and temporal subset parameters were defined in factbase, and finally, inference was executed.

High water count values can be visualized within the lake while 0 or low water count can be seen at the border of the lake in Figure 1. Likewise, pixel-wise cloud count and snow count can be effectively visualized in Figure 4 and Figure 5 respectively.

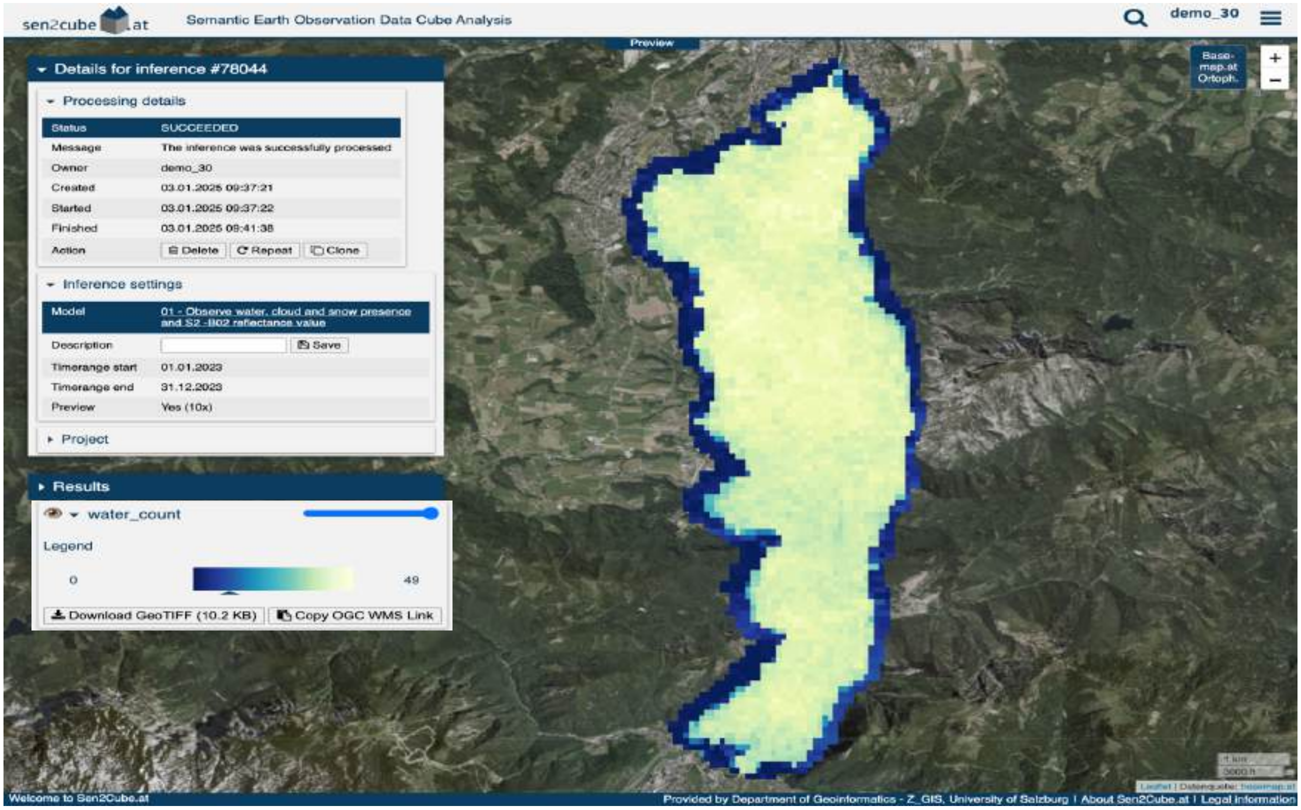


Figure 3: Traunsee lake water count map

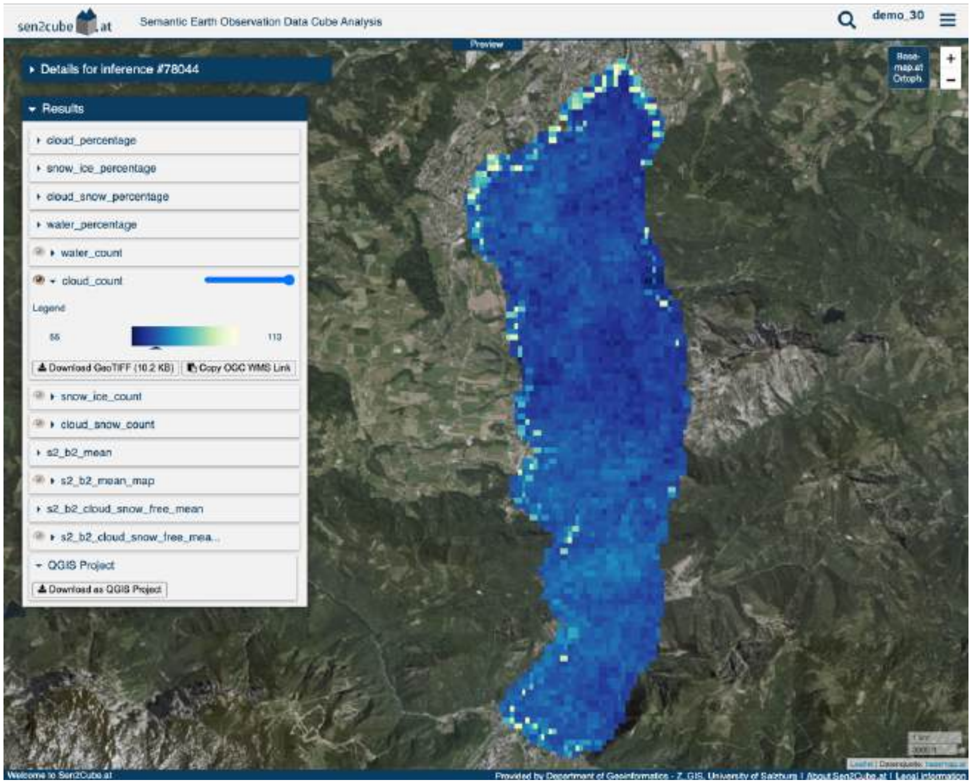


Figure 4: Traunsee lake cloud count map

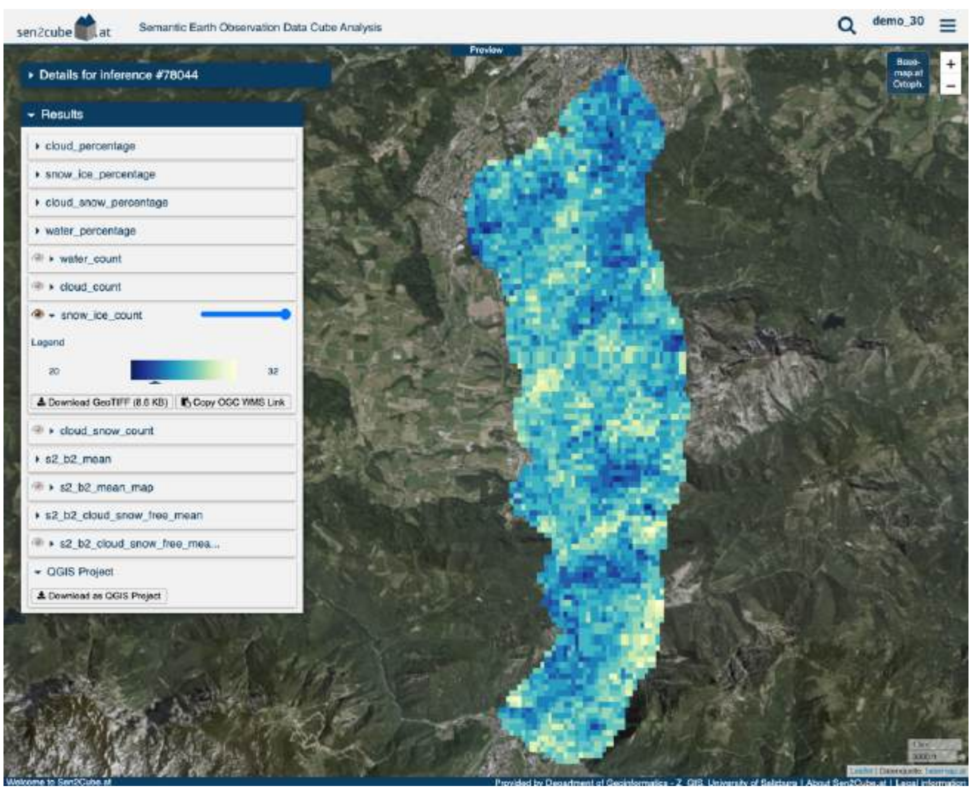


Figure 5: Traunsee lake snow/ice count map

Figure 6 and Figure 7 represent the cloud and snow/ice percentage coverage that is reduced over space respectively. The frequency of snow/ice coverage was high from January to May and in December.

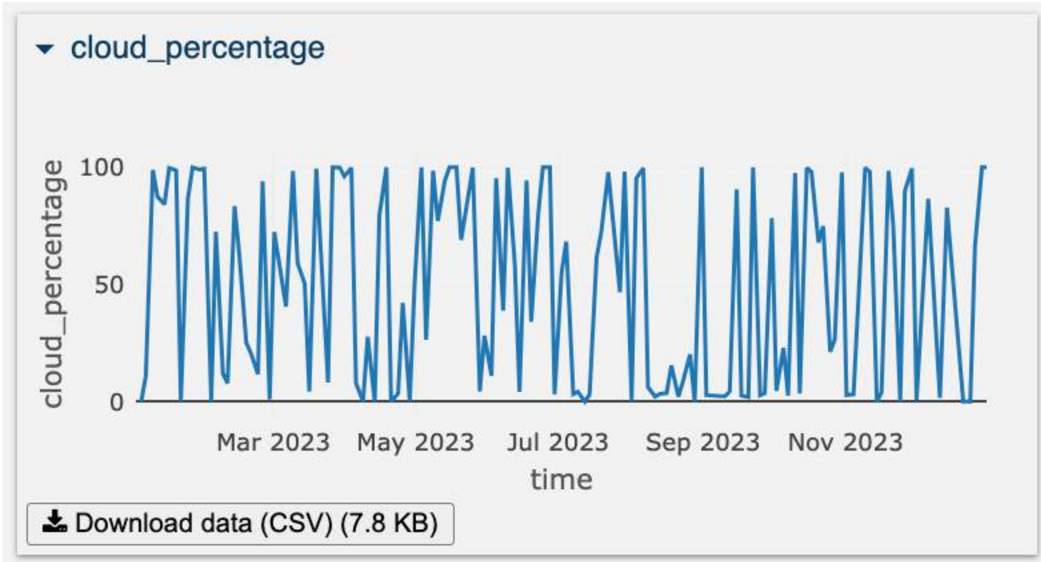


Figure 6: Traunsee lake cloud percentage graph

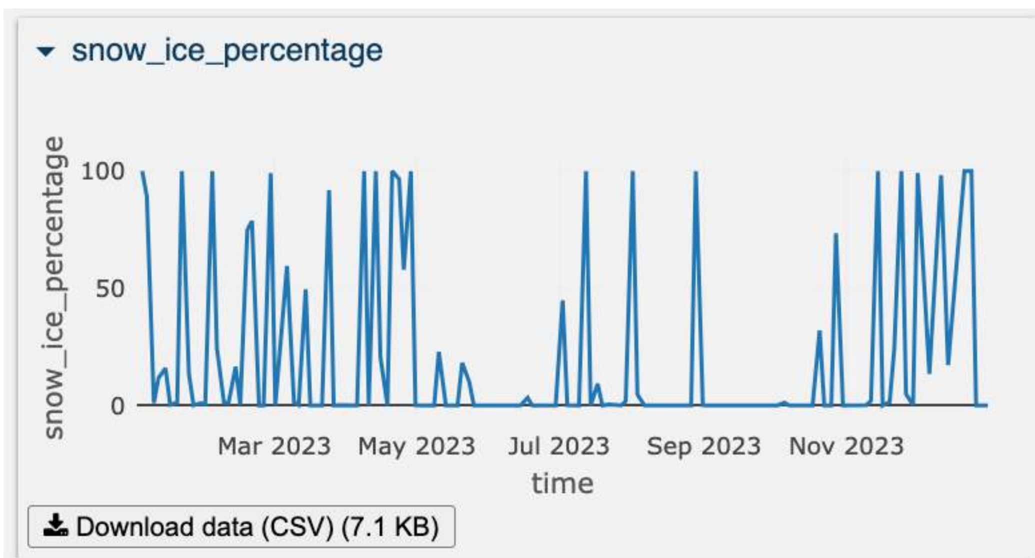


Figure 7: Traunsee lake snow/ice percentage graph

Figure 8 represents the combined cloud and snow/ice percentage graph reduction over space and Figure 9 represents the corresponding water percentage of the Traunsee lake for the year 2023. From the graphs, it can be depicted that whenever there is 100% cloud or snow coverage, the water percentage is zero, and when there is no cloud or snow, water percentage values are high which is obvious.

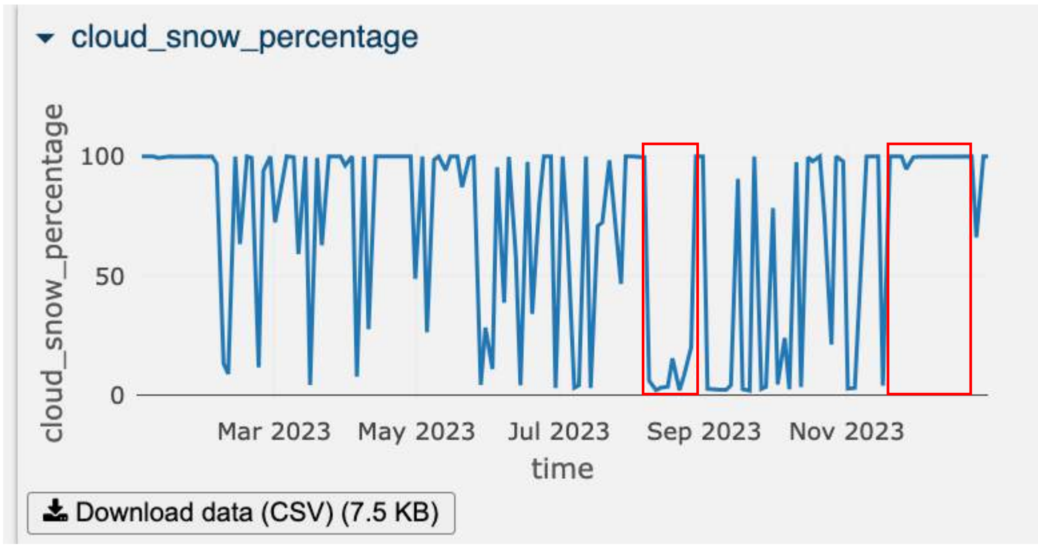


Figure 8: Traunsee lake cloud and snow percentage graph

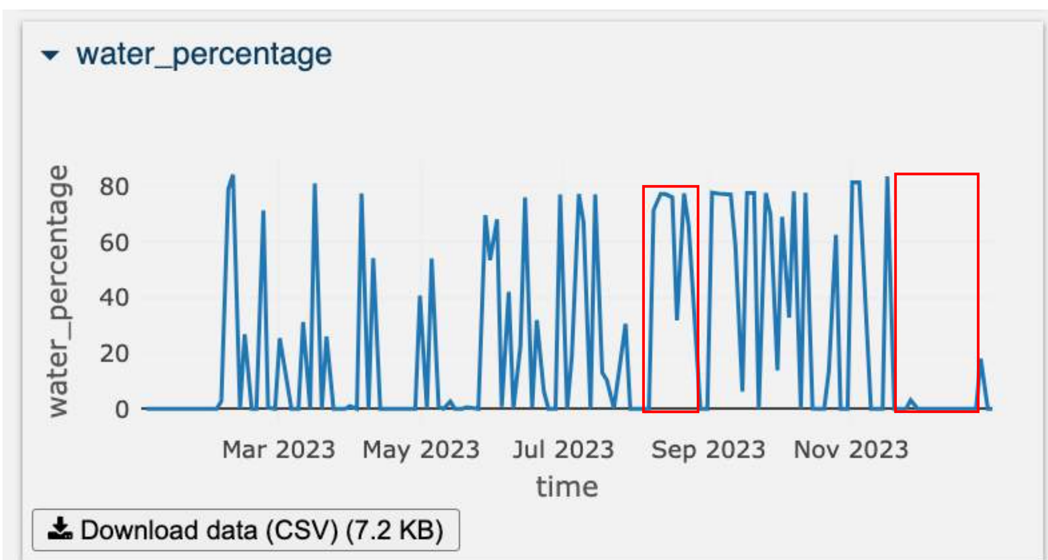


Figure 9: Traunsee lake water percentage graph

Figure 10 and Figure 11 represent sentinel 2 band 2 mean reflectance map reduced over time and sentinel 2 band 2 mean reflectance graph reduced over space respectively. Whereas Figure 13 and Figure 14 represent the corresponding cloud and snow/ice-free band 2 mean reflectance results. From the figure, it can be analyzed that reflectance is higher in the lower part of the lake compared to the upper part.

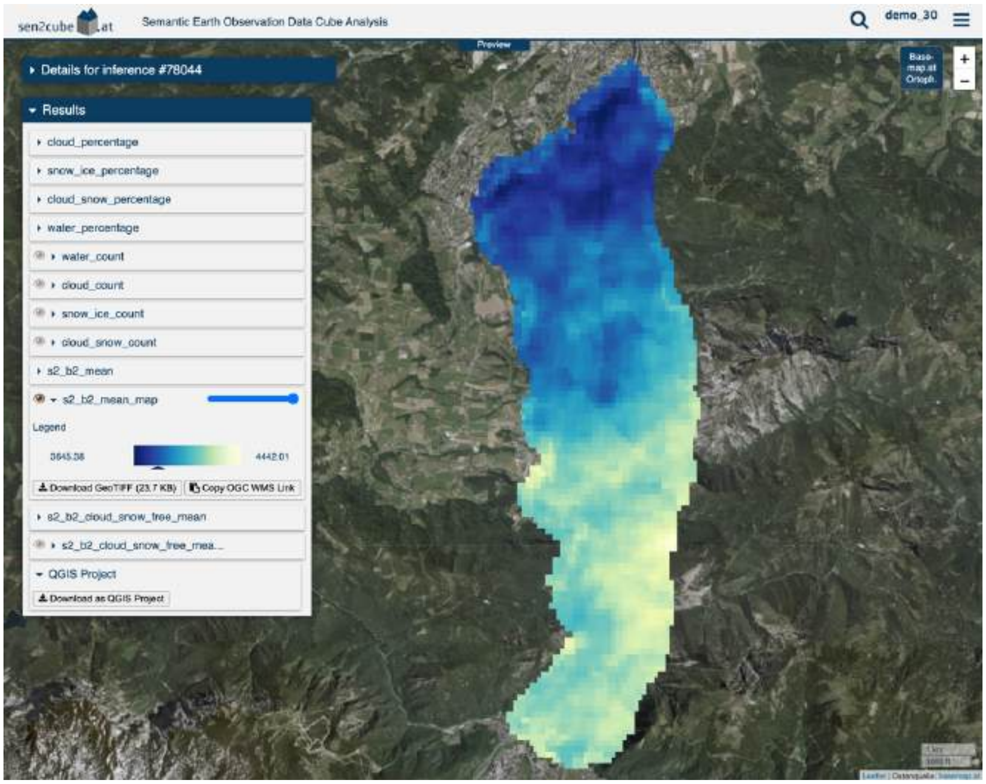


Figure 10: Traunsee lake blue band mean reflectance map

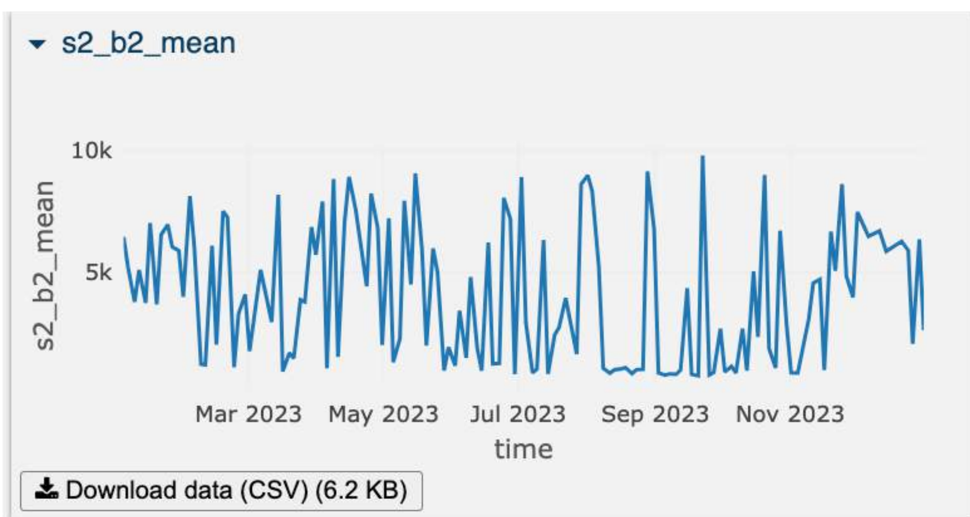


Figure 11: Traunsee lake blue band mean reflectance graph

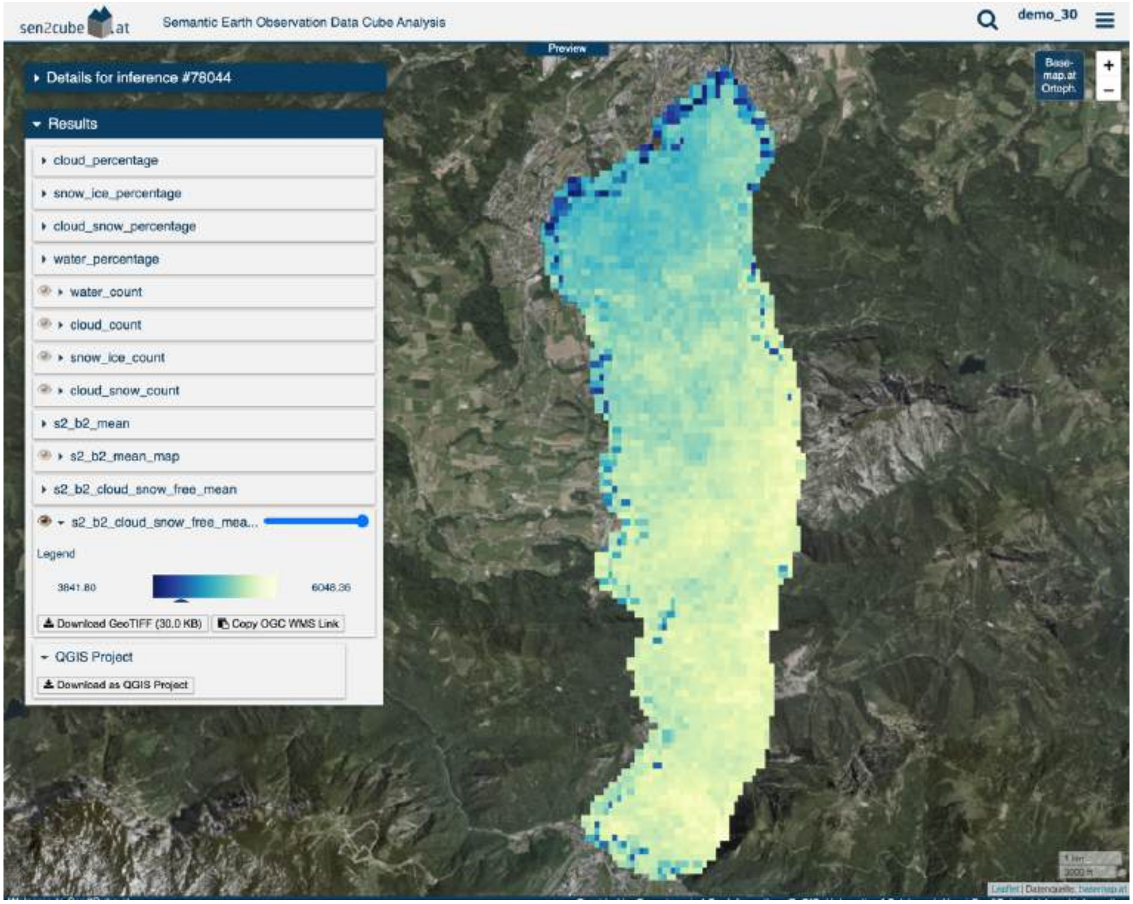


Figure 12: Traunsee Lake cloud and snow-free blue band mean reflectance map

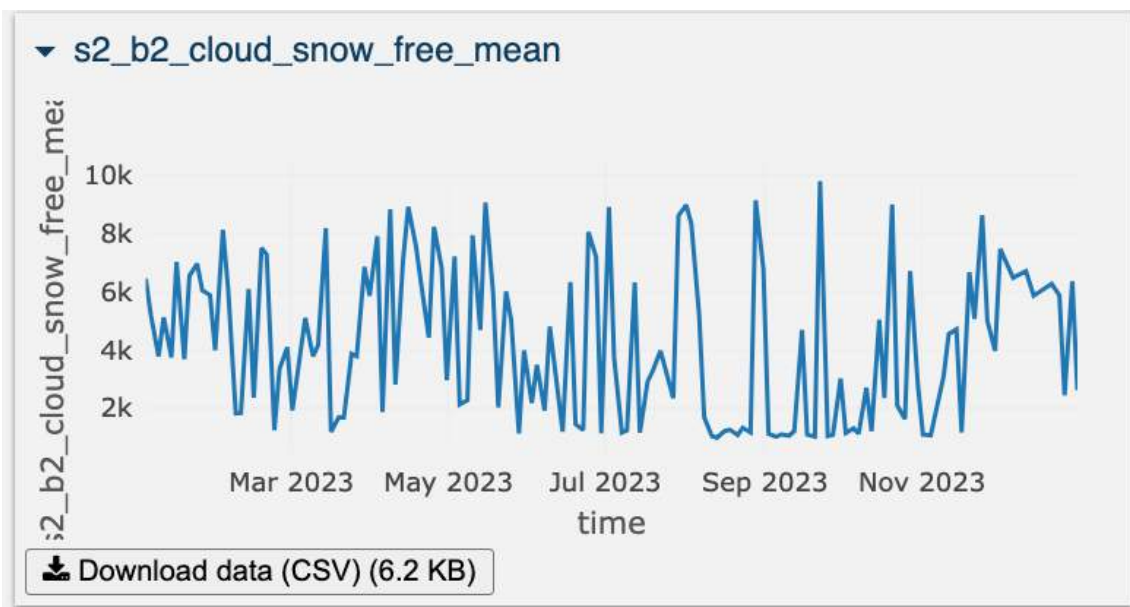


Figure 13: Traunsee Lake cloud and snow-free blue band mean reflectance graph

3.2 Millstatter Lake

Figure 14 represents the Millstatter Lake study area map which is also drawn free hand using polygon drawing tool. Model were developed and selected in the knowledgebase, spatial and temporal subset parameters were defined in factbase, and finally, inference was executed.

Figure 15, Figure 16, and Figure 17 represent water, cloud, and snow/ice count maps that is reduced over time respectively.

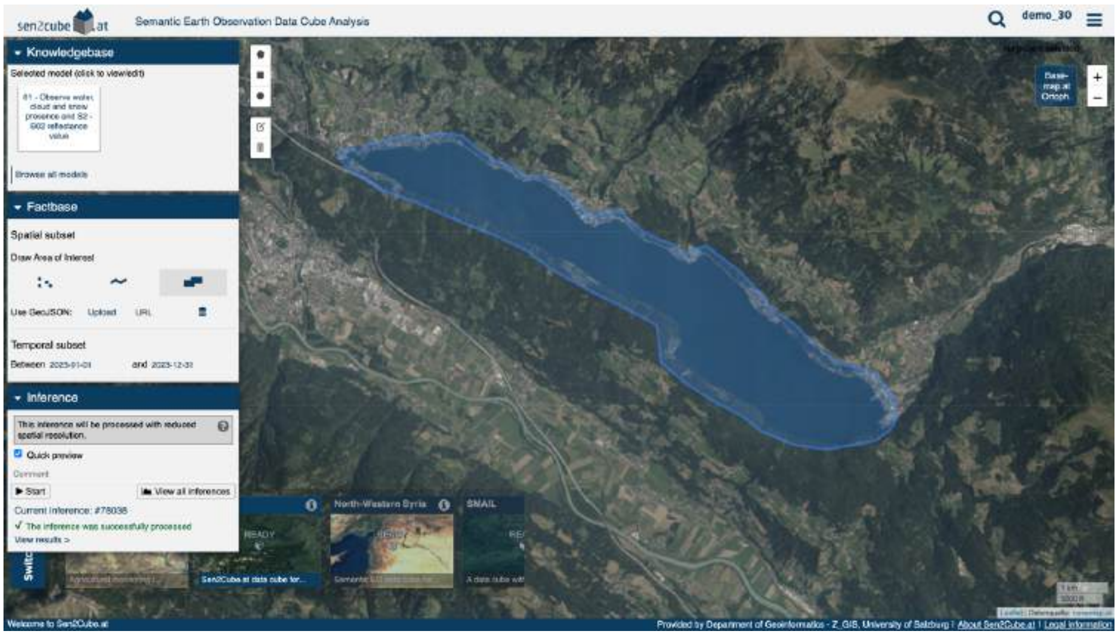


Figure 14: Millstatter lake study area inference setup

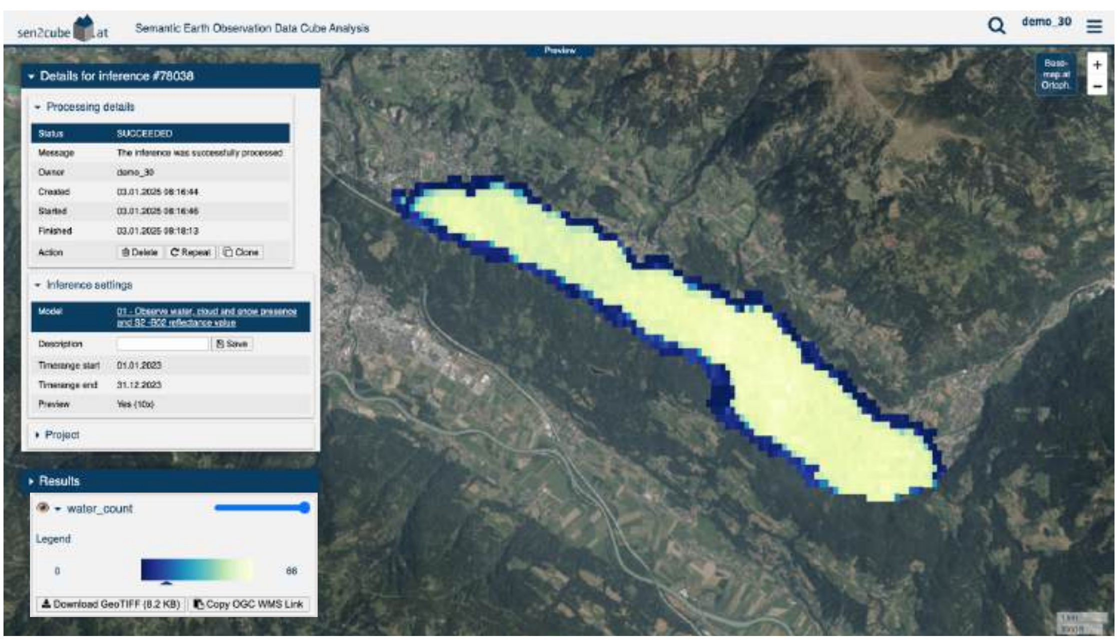


Figure 15: Millstatter lake water count map

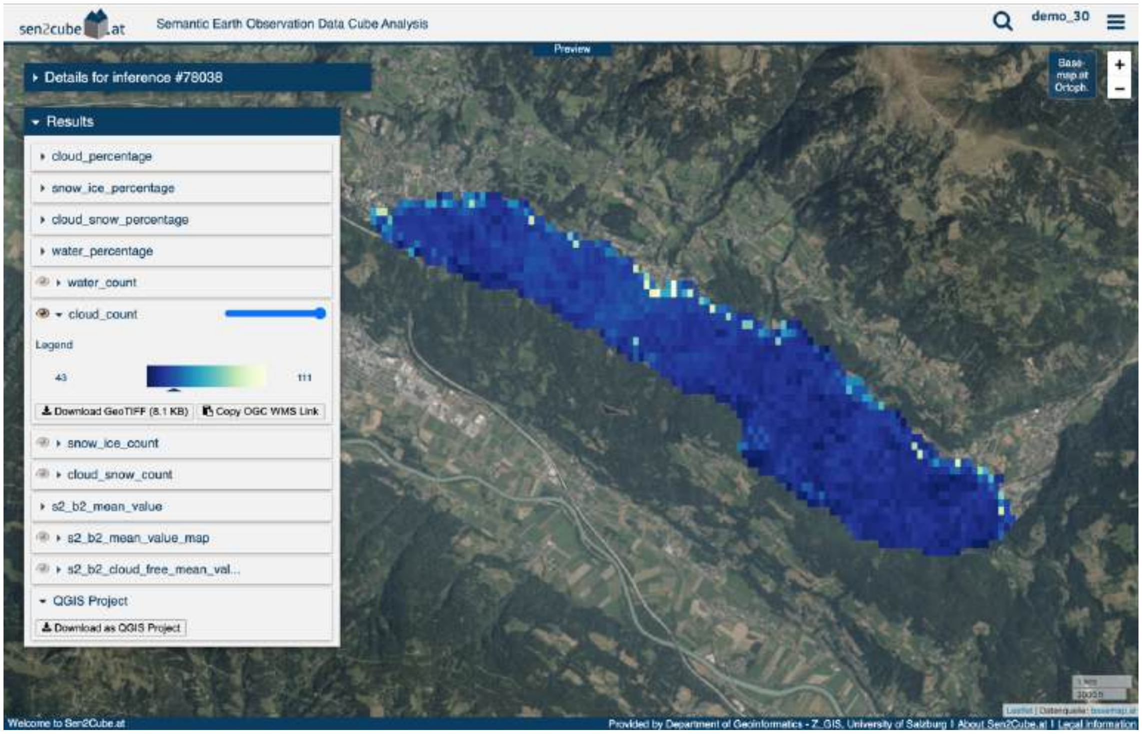


Figure 16: Millstatter lake cloud count map

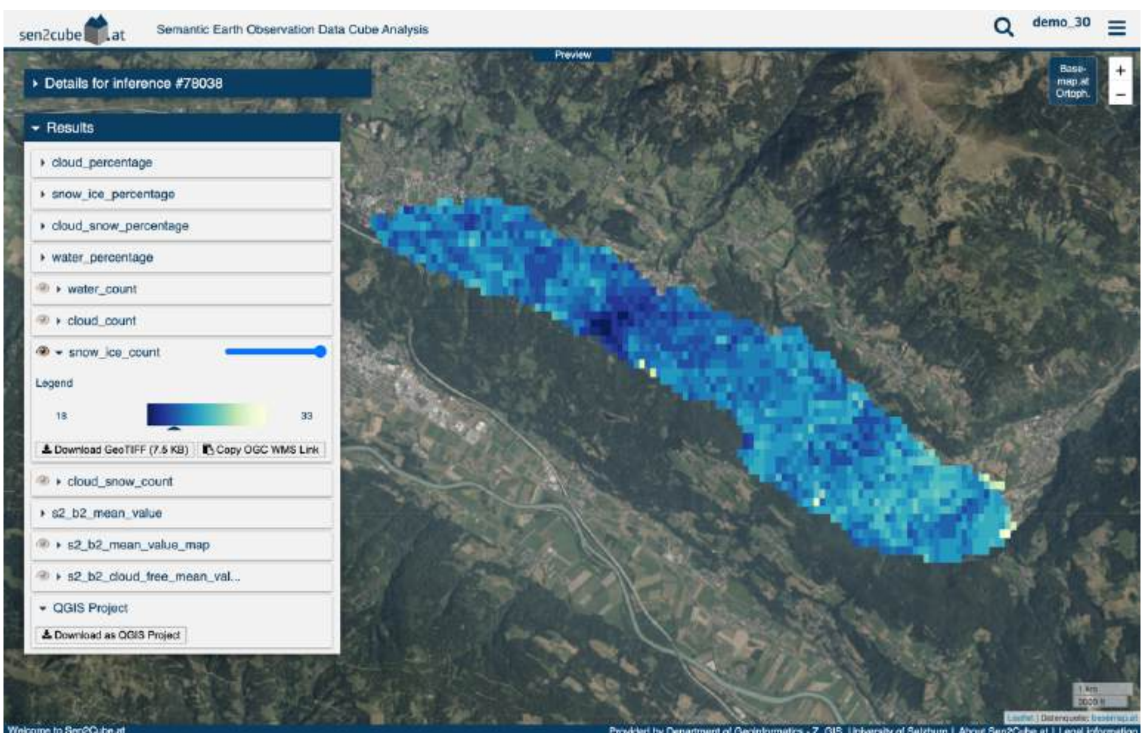


Figure 17: Millstatter lake snow/ice count map

Figure 19 depicts that Millstatter Lake experienced more snow/ice coverage during January, April, and May.

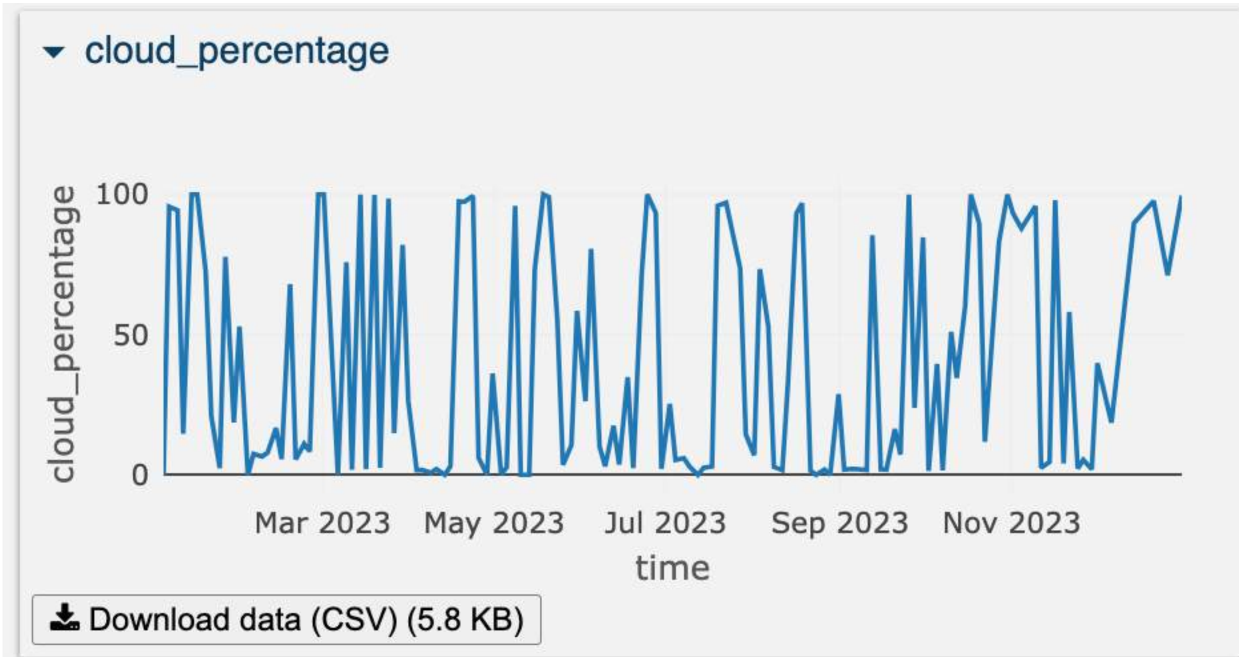


Figure 18: Millstatter lake cloud percentage graph

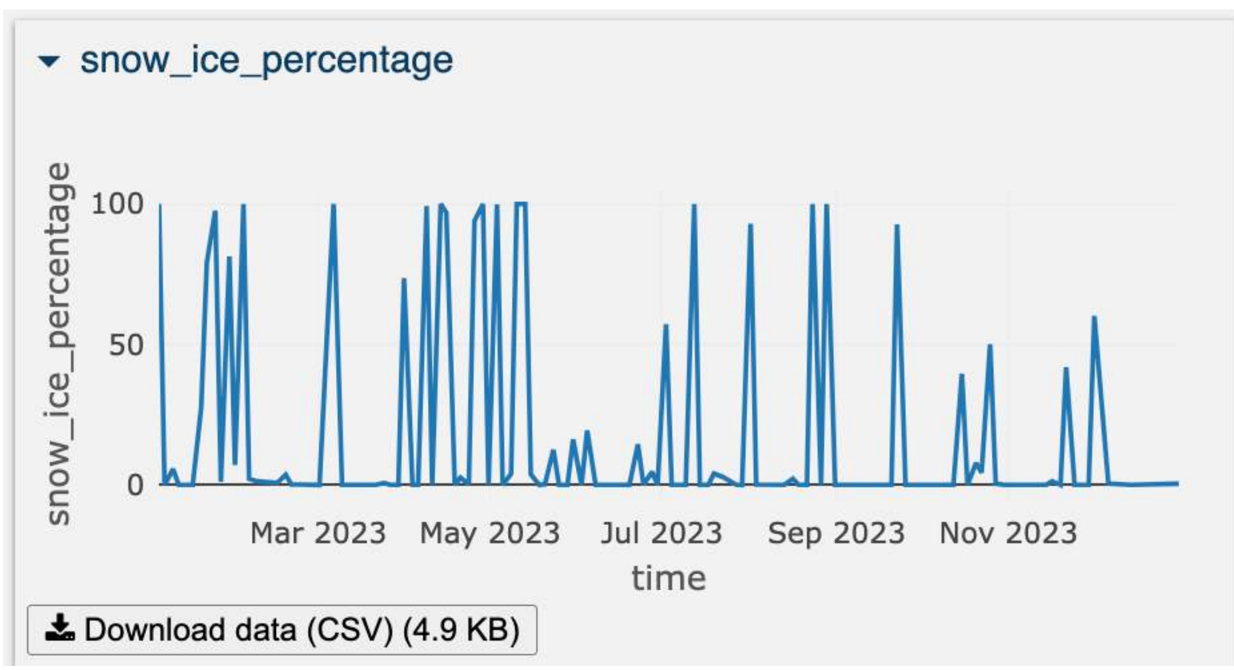


Figure 19: Millstatter lake snow/ice percentage graph

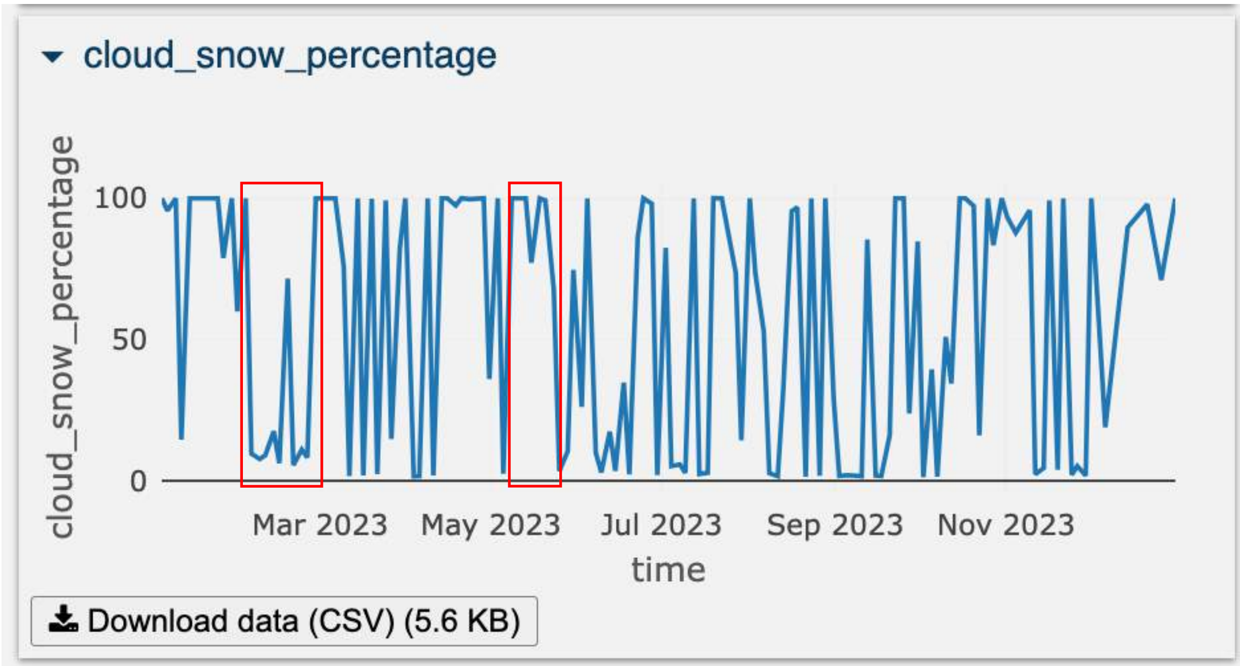


Figure 20: Millstatter Lake cloud and snow/ice percentage graph

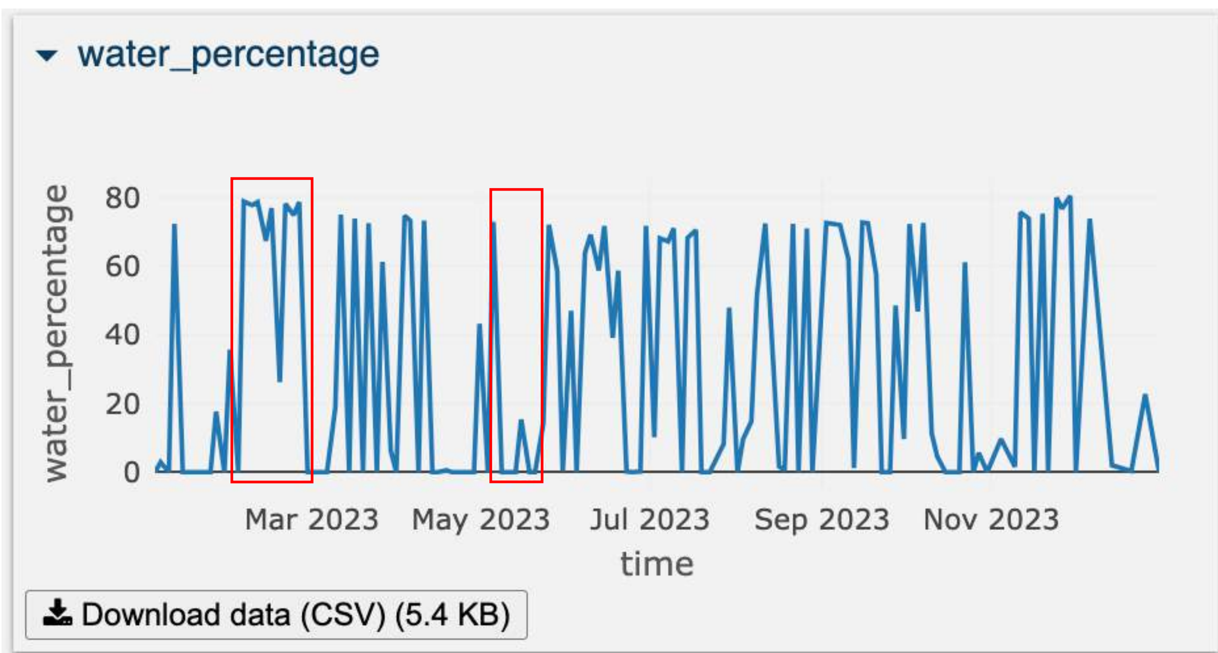


Figure 21: Millstatter Lake water percentage map

Figure 22 and Figure 23 represent sentinel 2 band 2 mean reflectance map reduced over time and the sentinel 2 band 2 mean reflectance graph reduced over space respectively. Whereas Figure 24 and Figure 25 represent the corresponding cloud and snow/ice-free band 2 mean reflectance results. From the figures, it can be analyzed that, unlike Traunseelake, Millstatter Lake had an overall uniform reflectance value. However, patches with lower reflectance values also can be seen.

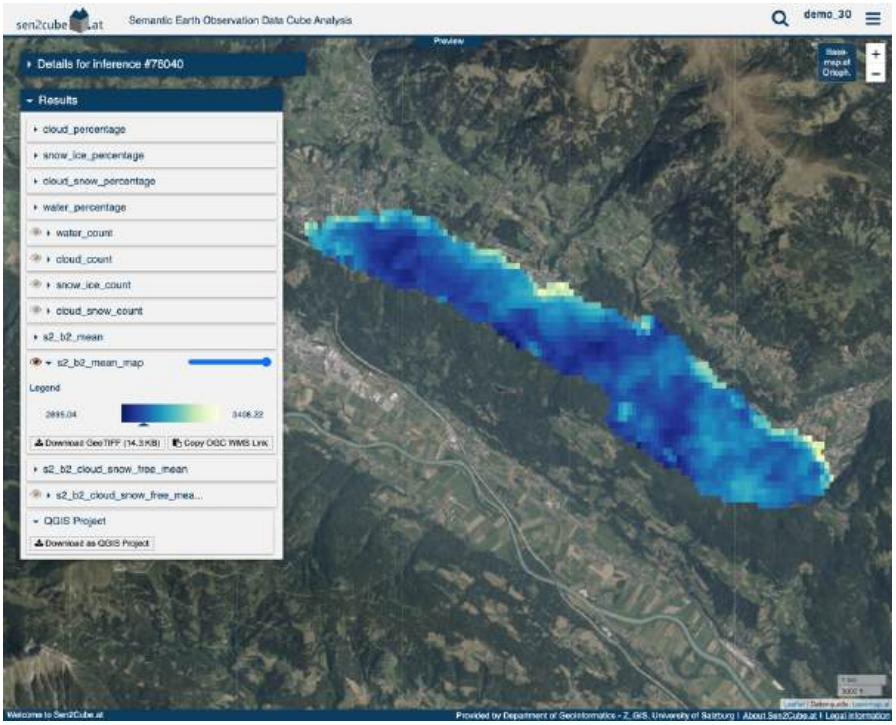


Figure 22: Millstatter Lake blue band mean reflectance map

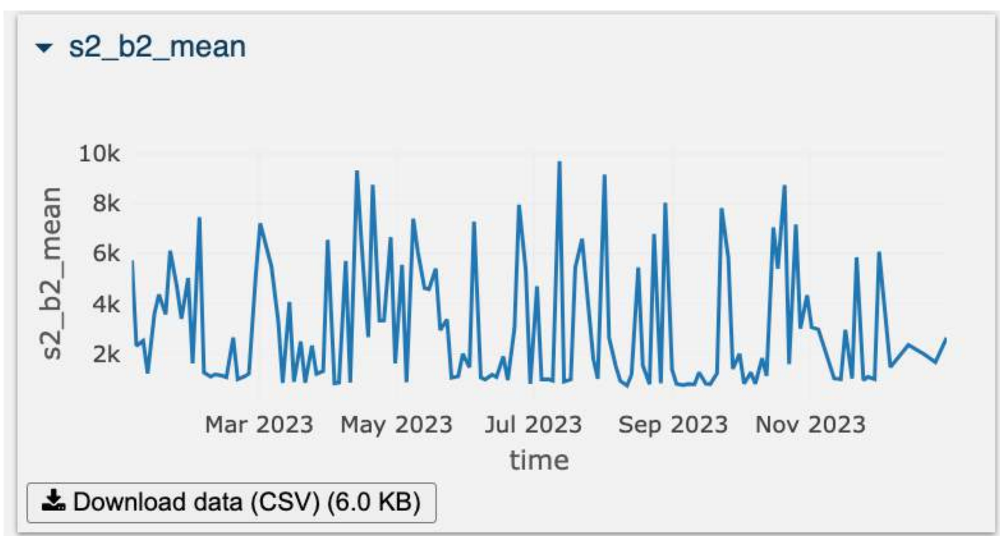


Figure 23: Millstatter Lake blue band mean reflectance graph

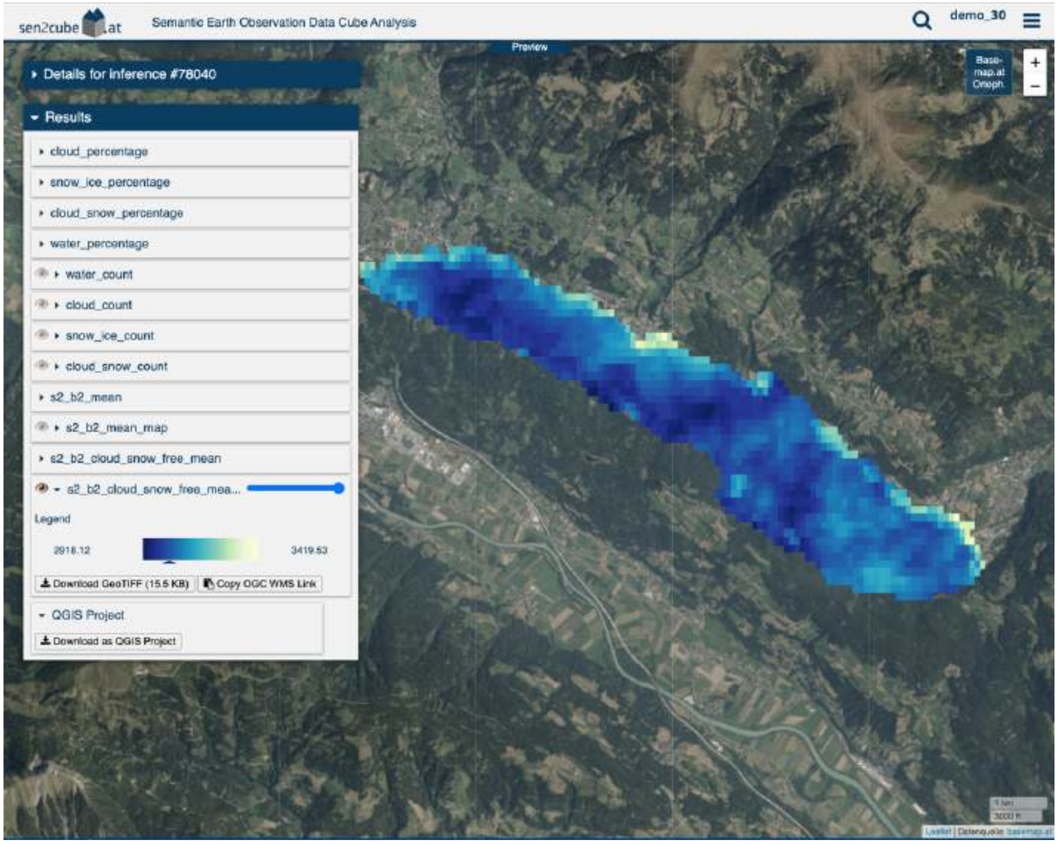


Figure 24: Millstatter Lake cloud and ice/esnow-free blue band mean reflectance map

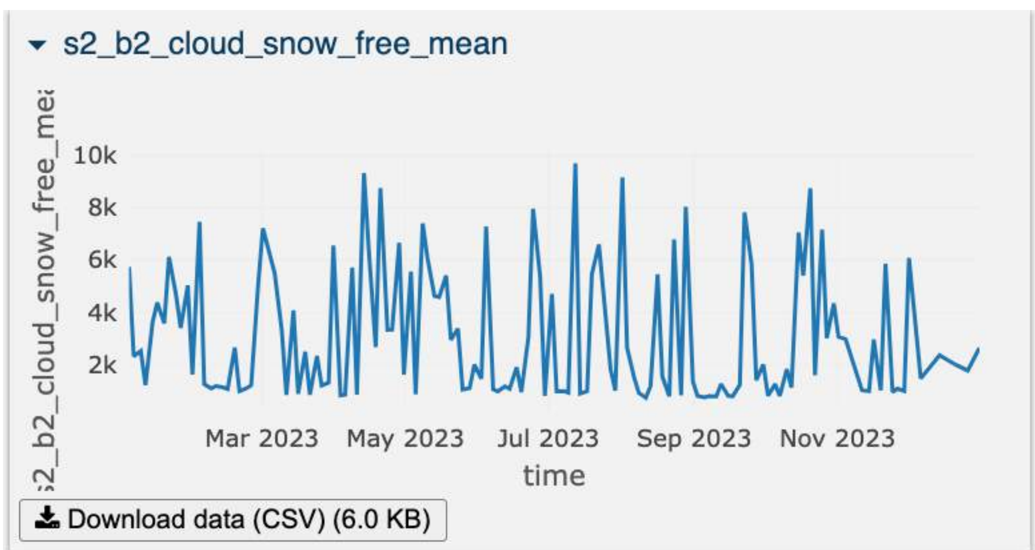


Figure 25: Millstatter Lake cloud and snow-free blue band mean reflectance graph

3.3 Neusiedl lake

Figure 26 represents the Neusiedl Lake study area map which is drawn freehand using a polygon draw tool. Model were developed and selected in knowledgebase, spatial and temporal subset parameters are defined in factbase, and finally inference was executed.

Figure 27, Figure 28, and Figure 29 represents water, cloud, and snow/ice count map that is reduced over time respectively.

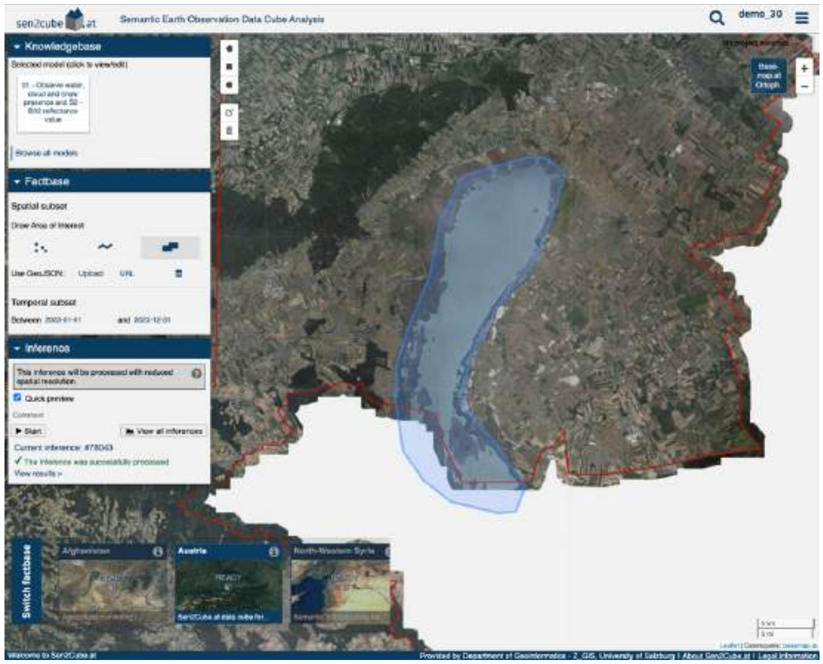


Figure 26: Neusiedl lake water count map

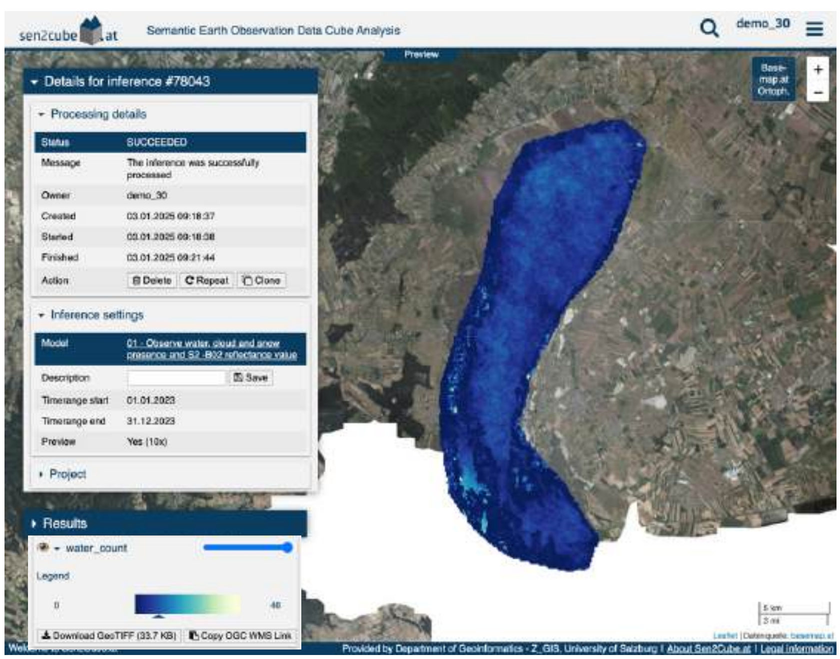


Figure 27: Neusiedl lake study area inference setup

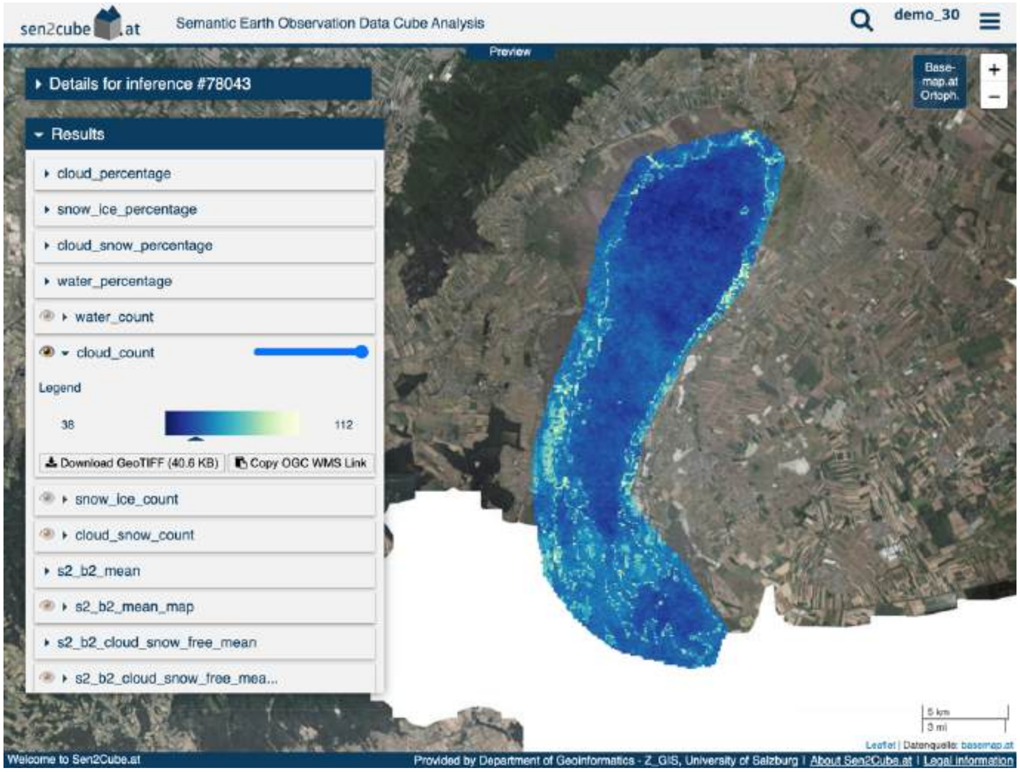


Figure 28: Neusiedl Lake cloud count map

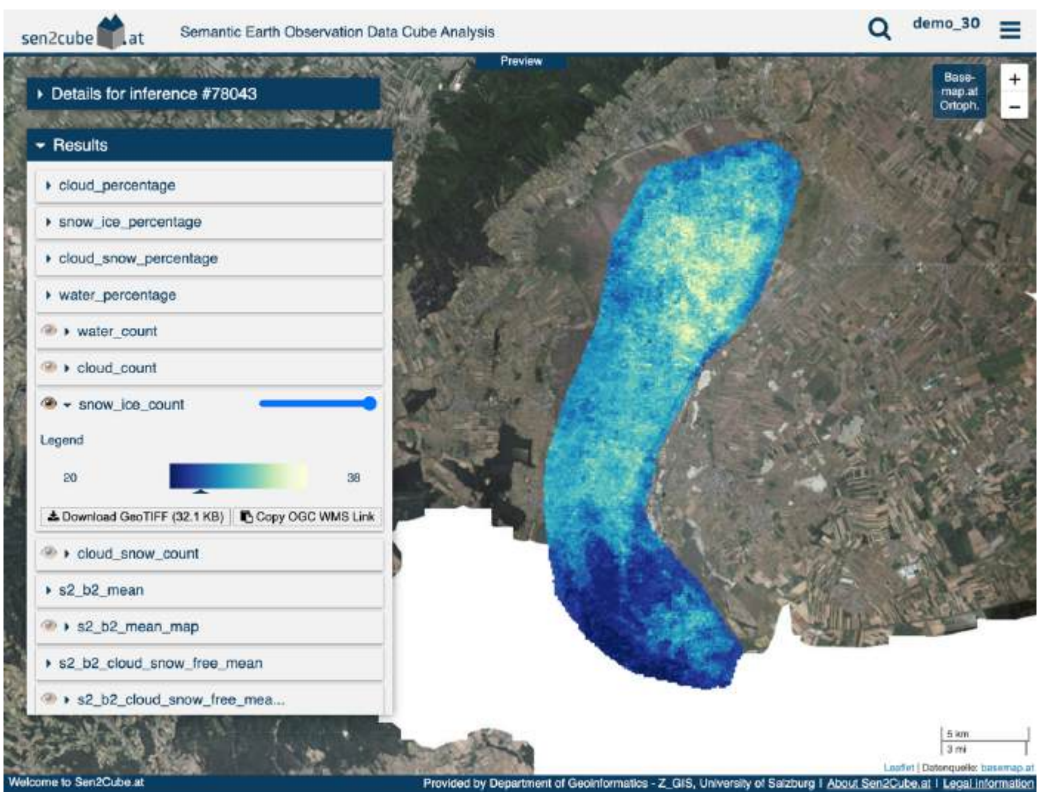


Figure 29: Neusiedl Lake snow/ice count map

From Figure 30 and Figure 31, it can be analyzed that snow/ice coverage was experienced almost every month in Neusiedl Lake.

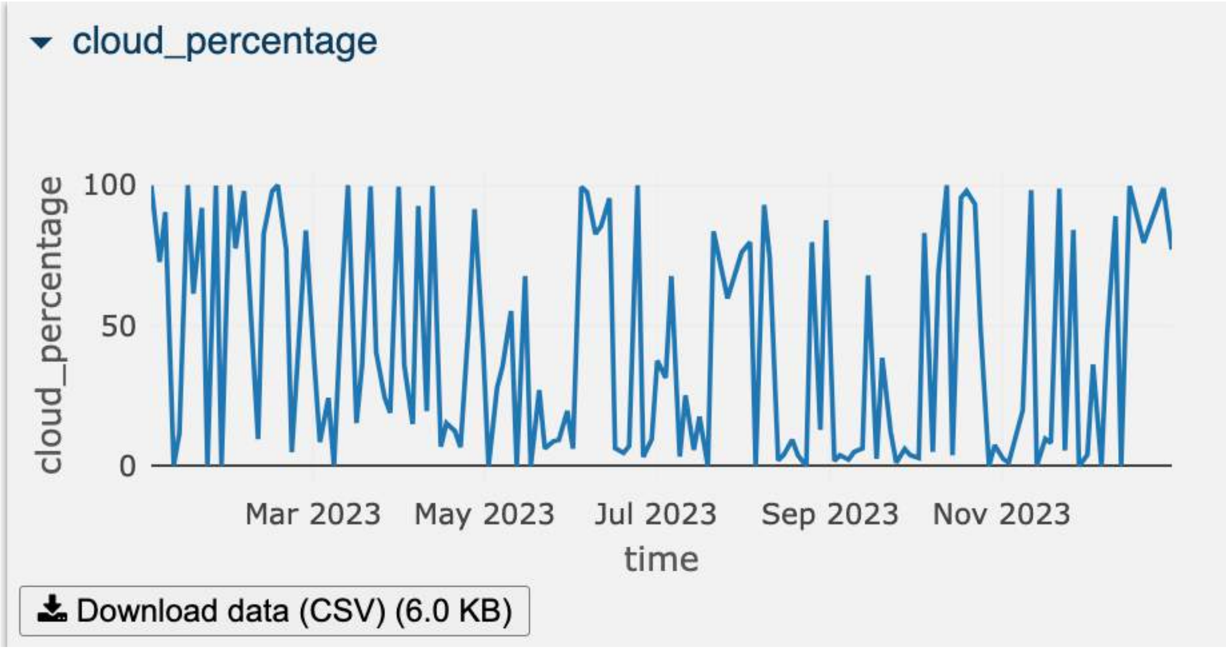


Figure 30: Neusiedl Lake cloud percentage graph

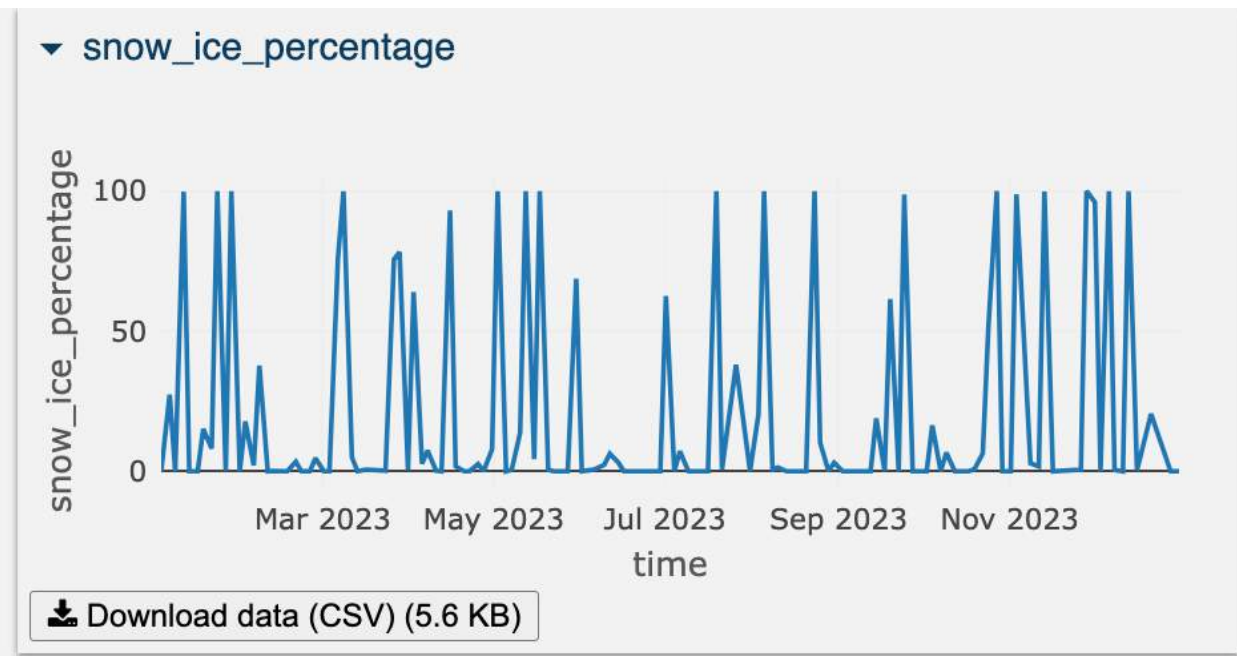


Figure 31: Neusiedl Lake snow/ice percentage graph

From the figure 32 and Figure 33, it can be analyzed that the water percentage throughout the year was relatively very low due to the high frequency of snow/ice coverage as well as the cloud coverage in Neuseidl Lake.

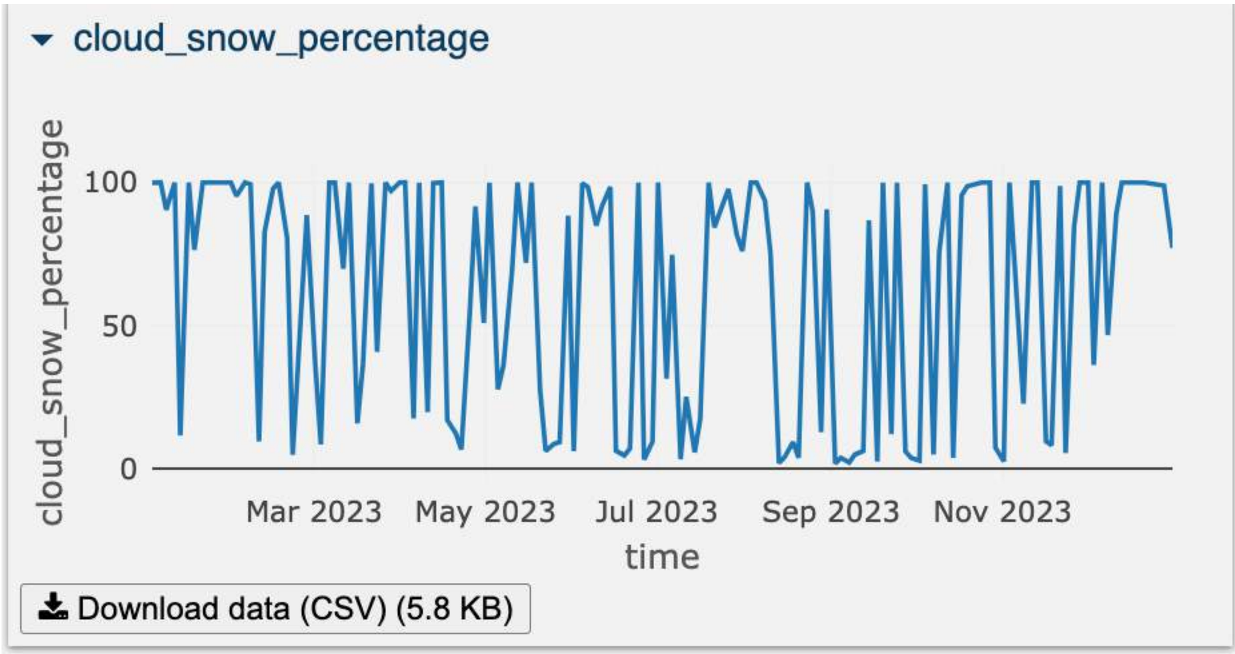


Figure 32: Neusiedl lake cloud and snow/ice percentage graph

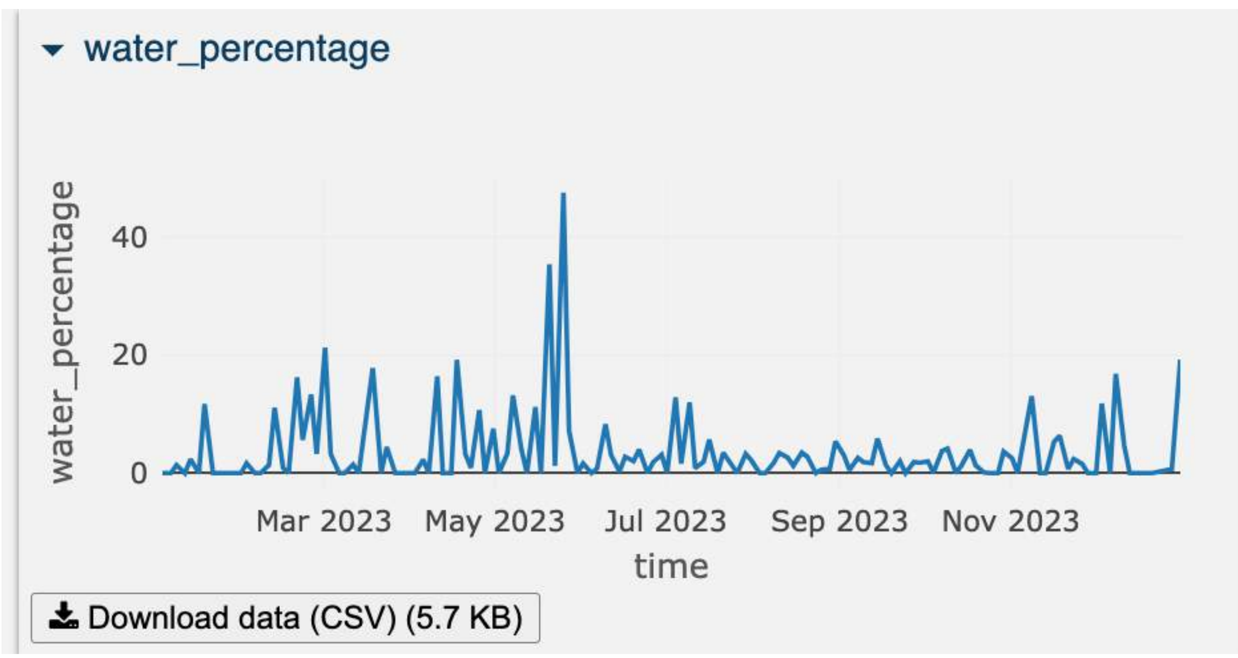


Figure 33: Neusiedl lake water percentage graph

Figure 34 and Figure 35 represent the sentinel 2 band 2 mean reflectance map reduced over time and sentinel 2 band 2 mean reflectance graph reduced over space respectively. Whereas Figure 36 and Figure 37 represent the corresponding cloud and snow/ice-free band 2 mean reflectance results. From the figures, it can be analyzed that Neusiedl lake had a lower band 2 reflectance value at the lower part of the lake in comparison to the upper part.

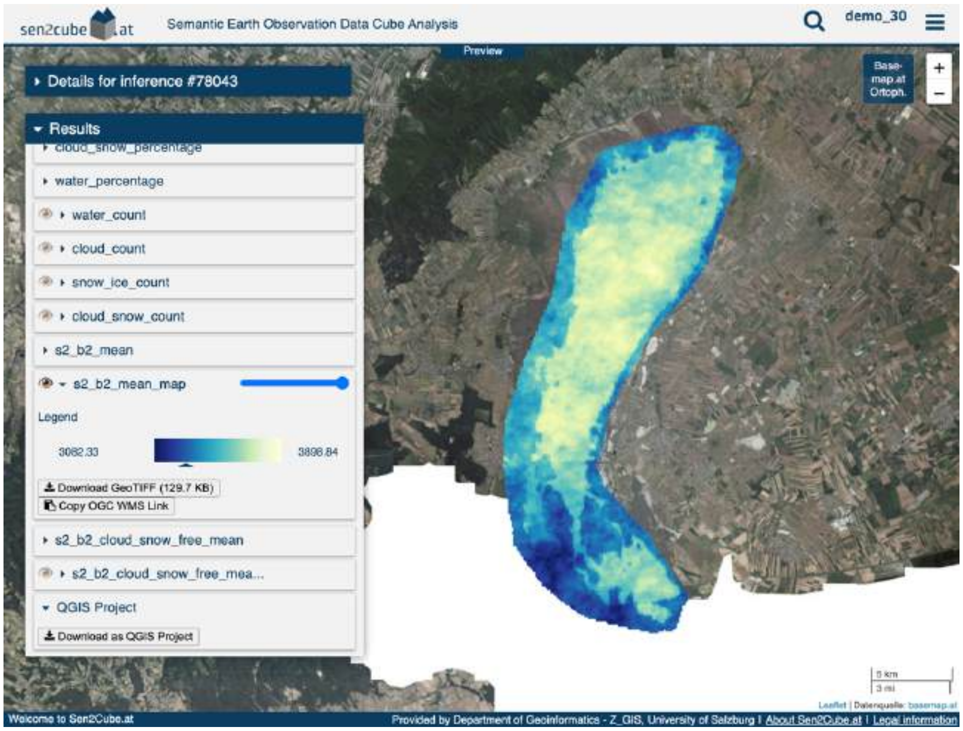


Figure 34: Neusiedl lake blue band mean reflectance map

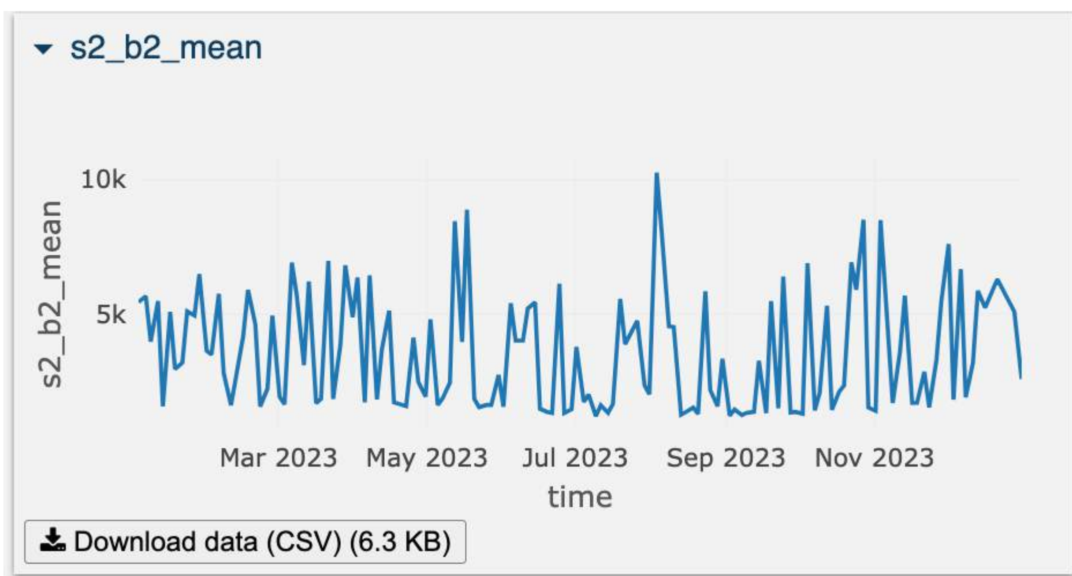


Figure 35: Neusiedl lake blue band mean reflectance graph

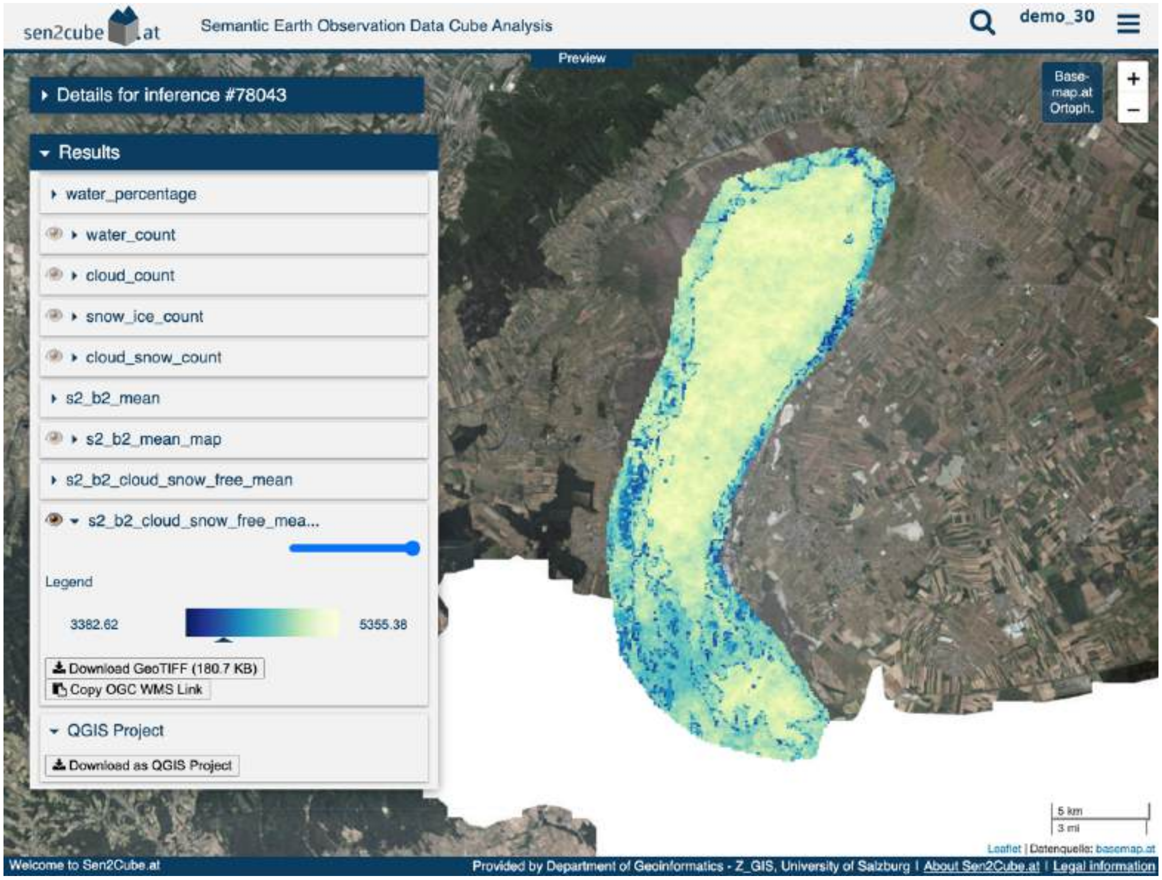


Figure 36: Neusiedl Lake cloud and snow free blue band mean reflectance map

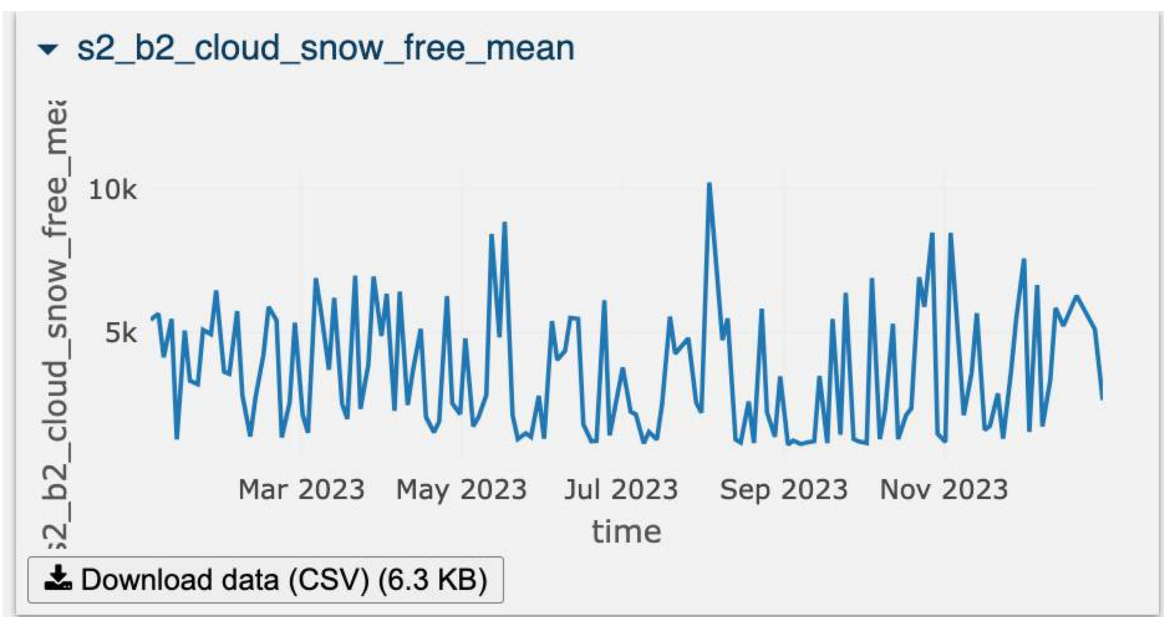


Figure 37: Neusiedl Lake cloud and snow-free blue band mean reflectance graph

4. Conclusion

To conclude, this assignment's purpose was to build a semantic-based model in an interactive no-code web interface of Sen2Cube by combining several building blocks. Varying characteristics and patterns of water, cloud coverage, snow/ice coverage, and blue band reflectance in three lakes of Austria throughout the year were analyzed effectively. An analysis based on reduction over space is very useful to make a temporal analysis for a defined interval of time, whereas an analysis based on reduction over time is useful to make a visualization in the form of a map that helps in understanding the spatial distribution of a variable.